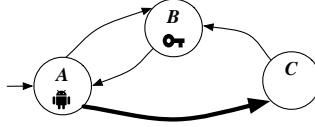


Part 1 - Artificial Intelligence (Time to complete Part 1: 100 minutes)

Consider the following scenario. A robot, *self*, can move across a network of locations shown in the picture below.



To traverse some edges of the network *self* needs a key (specifically the edge in **bold** in the picture).

Non-fluents predicates: $Location(x)$ denoting locations, in our case A, B, C ; $Obj(x)$ denoting the objects, in our case only the key *key*; $connected(x, y)$ where x and y are locations, denoting the edges not requiring a key in the network (see picture above); $connectedKey(x, y)$ where x and y are locations, denoting the edges that need the key to be traversed (see the picture above).

Fluents: $SelfAt(x)$ where x is a location, denoting that *self* is at location x ; $At(x, y)$ where x is an object and y a location, denoting that the object x is at location y ; $SelfHas(x)$ where x is an object, and denotes that *self* has object x with himself/herself.

Actions as available to *self*:

- $goto(\ell_1, \ell_2)$ to move from location ℓ_1 to location ℓ_2 . To do so *self* needs to be in location ℓ_1 , moreover if traversing the edge in the network needs the key, *self* need to have the key. The effect of $goto(\ell_1, \ell_2)$ is that *the new location of self*, together with all the objects that *self* has with him/her (i.e., the key), will be ℓ_2 .
- $pickup(obj)$ to pick up object *obj* (i.e., the key). To do so *self* needs to be in the same location as *obj*. The effect of $pickup(obj)$ is that *self* will have *obj* with himself/herself.
- ~~$drop(obj)$ to drop *obj* (i.e., the key). To do so *self* needs to have *obj* with himself/herself. The effect of $drop(obj)$ is that *self* will not have *obj* with himself/herself and the location of *obj* will be that of *self*.~~

Initially: *self* is at location A , the key is at location B , and *self* does not have any object with himself/herself. (The non-fluent predicates have their extension as above.)

Exercise 1. Formalize the above scenario as a Basic Action Theory in Reiter's Situation Calculus. Then check by regression:

1. Whether the action $goto(A, C)$ is executable in S_0 ;
2. Whether the sequence of actions $goto(A, B); pickup(key); goto(B, A); goto(A, C)$ is executable in S_0 ;
3. Whether, after executing the sequence of actions $goto(A, B); pickup(key); goto(B, A); goto(A, C)$ in S_0 , *self* has the key with himself/herself.

Exercise 2. Considering as goal $SelfAt(C)$, formalize the above scenario as a PDDL domain file and PDDL problem file. Then:

1. Draw the corresponding transition system;
2. Solve planning for achieving the above goal by using backward fixpoint computation, reporting the various stages of the fixpoint computation;
3. Solve planning for achieving the above goal by using forward depth-first search (uniformed), reporting the various steps of the forward search computation.

Exercise 3. Check, by using tableaux, whether the following FOL formula is indeed valid, and if not show a counterexample:
 $(\forall x.(A(x) \supset B(x))) \equiv ((\forall y.A(y)) \supset (\forall z.B(z)))$.

Exercise 4. Describe the DPLL procedure for satisfiability and use it to find a model (a satisfying interpretation) of the following set of clauses: $\{ \{A, B, \neg C\}, \{\neg A, B, C\}, \{\neg A\}, \{\neg B, C\} \}$.

SAPIENZA Università di Roma–MSc. in Engineering in Computer Science

Artificial Intelligence and Machine Learning

Part 2: Machine Learning

23 June 2022

- **Time Limits:** AI+ML: 2h 30m; AI only: 1h 40m; ML only: 50m.
- **Important:** The AI and ML parts must be delivered separately (one set of sheets for each part). Write your name and matricola number on every delivered sheet.

Exercise 5

You need to devise a system for oil-price prediction. The available data include N instances, each representing a sequence of average prices for 11 consecutive days. That is, the generic instance has the form $x = (d_1, \dots, d_{11})$, where d_i represents the average oil price of the i -th day in the sequence.

1. Design a neural network that can predict the average price of one day, given that of the previous consecutive 10 days. The ANN must include two hidden layers (plus one output layer). The first hidden layer contains 30 units. The design must specify: the dimensions of the weight matrices W_1 , W_2 and W_3 of the two hidden and the output layers, respectively, and the type of units used in each layer. Bias weights can be ignored.
2. Given the design above, indicate the number of units of the second hidden and the output layers, and the number of trainable parameters of the network.
3. Indicate one plausible algorithm for training the ANN (no need to report the pseudo-code) and explain how you would use the available N instances in the dataset to evaluate the performance of the system.

Exercise 6

1. Define the notion of Markov Decision Process (MDP). For each element of an MDP, provide its mathematical definition and an intuitive description (what they model). Indicate which elements are unknown in Reinforcement Learning.
2. Illustrate and explain the *Markov Assumption*.
3. Define the notions of (*expected*) *cumulative discounted reward* and *optimal policy*.